Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
		A ROTOR				
1	NS6-100	Blade, glass/epoxy, glass UD caps (B1–B3)	3	make	E-glass/epoxy, PVC core	20.6
2	NS6-105	Root insert M12 (10 per blade, bonded)	30	make	C45E+QT zinc-flake coated	0.3
3	NS6-130	Pitch washer plates 28 × 4 hardened (30)	1	make	C45 hardened 45 HRC, zinc-flake	0.5
4	NS6-200	Fixed hub weldment	1	make	S355NL, welded, stress relieved, machined	79.2
5	NS6-210	Spinner (GRP, 4 mm)	1	make	glass/polyester, gelcoat RAL 9010	8.7
6	NS6-211	Spinner brackets (3)	1	make	S235 galv.	2.0
		B DRIVETRAIN AND GENERATOR G6				
7	NP1-100	Spindle	1	make	42CrMo4+QT forged	56.5
8	NP1-101	Front spacer / seal counterface Ø180 (Rev B)	1	make	42CrMo4+QT, induction hardened	1.4
9	NP1-102	Rear seal sleeve Ø195 on the Ø155 shoulder (Rev B)	1	make	42CrMo4+QT, induction hardened	1.3
10	NP1-200C	Stator carrier with 45 dovetail grooves	1	make		39.9
11	NP1-201C	Stator: 45 tooth-segment lamination stacks	1	make		50.5
12	NP1-202	Stator front clamp ring	1	make	1.4404 (non-magnetic)	2.2
13	NP1-203C	Coils on segments, 45 × 28 turns (end windings shown)	1	make	Cu enamelled Ø1.55 class 200, NKN 0.30	20.0
14	NP1-210C	Rotor drum weldment (G6: no fins, no ribs)	1	make	S355J2+N	91.2
15	NP1-211-F	Front bearing seal cover	1	make	S355J2	1.5
16	NP1-211-R	Rear bearing seal cover, bore Ø198 (Rev B)	1	make	S355J2	1.4
17	NP1-220E	Magnets 200 × N45SH 30×15×12 + comb + sleeve	1	make	NdFeB N45SH	10.7
18	NP1-230	Brake disc Ø880/Ø652 with 12 lock holes (Rev B)	1	make	C45E+N 16 mm	33.6
19	NP1-310	Rotor brake caliper bracket (left, Rev B)	1	make	S355J2+N 25 plate	18.1
20	NP1-320	Rotor lock pin Ø20 (engaged, top)	1	make	42CrMo4+QT	0.5
21	NP1-321	Lock pin holder with 2 overspeed sensor seats	1	make	S355J2	24.5
22	NS6-610	G6 plain rear cover (no fan blades)	1	make	EN AW-5083-H111	7.2
23	NS6-913	Tapered roller bearing 32028 X (front + rear)	2	buy	bought 32028 X	4.0
24	NS6-914	Overspeed sensors M18 inductive (2, bought)	1	buy	bought M18 PNP	0.4
25	NS6-915	Spring-applied brake caliper, 1.5 kNm pack (bought)	1	buy	bought, spring-applied	9.0
		C NACELLE				
26	NS6-300	Bedplate weldment 6 kW (light)	1	make	S355J2+N plates, stress relieved, machined	122.1
27	NS6-700-1	Nacelle cover, lower tray (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	15.7
28	NS6-700-2	Nacelle cover, upper lid (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	17.5
29	NS6-710	Cover support arms + rear rail	1	make	S235 galv.	8.3
30	NS6-720	Met mast: anemometer, vane, air terminal	1	make	1.4404 tube 42.4 × 2 + bought sensors	4.5
		D YAW SYSTEM				
31	NP1-420	Yaw drive bracket	2	make	S355J2+N	7.7
32	NS6-911	Yaw slewing ring Dpw 360, z150 m4	1	buy	bought: 42CrMo4, induction hardened teeth	55.0
33	NS6-912	Yaw gearmotor 0.18 kW 1:1680, motor brake 2 Nm	2	buy	bought	19.0
34	NY-400	Tower-top adapter with friction track	1	make	S355J2+N weldment	106.7
35	NY-410	Friction pad unit	2	make	S355 / DIN 2093 springs / organic pads	14.0
36	NY-430	Yaw lock (screw block)	1	make	42CrMo4 / S355	6.5
37	NY-440	Yaw encoder + cam limit switch unit	1	make	S355 + absolute encoder + cam switch	6.0
38	NY-460	Cable hanger tube + loop cleat	1	make	S355 tube 168.3 × 4.5	10.0
		E ELECTRICAL				
39	NS6-600	Braking resistor box 3 × 4 Ω (vented)	1	make	1.4301 box + steel-grid resistors	18.0
40	NS6-620	Tower base cabinet IP55 800 × 600 × 300 on stand	1	make	stand S235 galv.; bought enclosure	95.0
41	NS6-730	Nacelle junction box IP66 (400 × 340 × 150)	1	make	bought: 1.4301 enclosure	9.0
42	NS6-800	Drop cable + base duct (cabling route)	1	make	H07RN-F 4G10 + 2 × LiYCY signal; PE duct Ø11	12.0
		F TOWER				
43	NS6-400	Tower 8 m, Ø460 → Ø320, t 10/8, one piece	1	make	S355J2+N, hot-dip galvanised	697.5
44	NS6-401	Hand-hole cover 200 × 350 × 6 with gasket	1	make	S355 galv.	3.3
45	NS6-410	Base hinge brackets + hinge pin Ø50	1	make	S355J2+N galv.; pin 42CrMo4+QT	37.4
		G FOUNDATION				

NOTES

1. Masses from the CAD model (cad/turbine_6kw/model6.py) for made parts; supplier data or estimates for bought parts. Full BOM with sources: bom6.csv.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BILL OF MATERIALS (1/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-002	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked